KAIST

AX20002 Introduction to AI model and algorithms

Homework 1: Regression Beyond the Gaussian Assumption

Lukas Lindholm | 20266508

Turned in: October 10, 2026

Contents

1 Introduction & General Methodology **3**

2 Task A **4**

3 Task B **7**

4 Task C **9**

5 Task D **11**

6 Conclusion **13**

1 Introduction & General Methodology

This assignment consisted of four distinct regression tasks, each with their own departure from the Gaussian assumption. Each task started by diagnosing the specific traits and attributes of the dataset and the deviations from the Gaussian baseline in order to decide on the correct modeling approach. This was done by studying plots to gain basic knowledge about the data and the characteristics of the residuals from the OLS regression model.

In order to best capture the severe non-linear relationships present in the data, the methodology chosen for regression was machine learning. More specifically, the gradient boosting frameworks `LightGBM (LGBM)` and `XGBoost` were utilized to learn each quantile $q \in \{0.1, 0.2, \ldots, 0.9\}$ directly. This was done by optimizing the Pinball loss function $\rho_q(u) = \max[qu, (q-1)u]$, $u = y - \hat{y}$, which is derived from performing MLE under the assumption that the residuals follow an Asymmetric Laplace Distribution. Through feature engineering using the insight from the plots, parametric optimization and safeguards for overfitting, the models adapted to the complex features of the data with satisfactory accuracy. To evaluate performance, a local 80/20 train-validation split was implemented, and the resulting scores consistently aligned with occasional submissions to the Kaggle public leaderboard, indicating that the models successfully generalized to unseen data without overfitting.

AI Usage Declaration The overarching strategic direction and algorithmic choices in this project were guided by an AI assistant (Google Gemini). The primary role of the student was that of a critical overseer: rigorously verifying the implementation, questioning the AI's suggestions, ensuring the logical soundness of the selected approaches, and evaluating alternative strategies. This dynamic allowed for an efficient exploration of machine learning concepts while maintaining strict human quality control and domain validation.

2 Task A

For task A, both the diagnostics plot of Figure 1 and the residuals vs. features plot in Figure 2 provided significant insights for tuning the model.

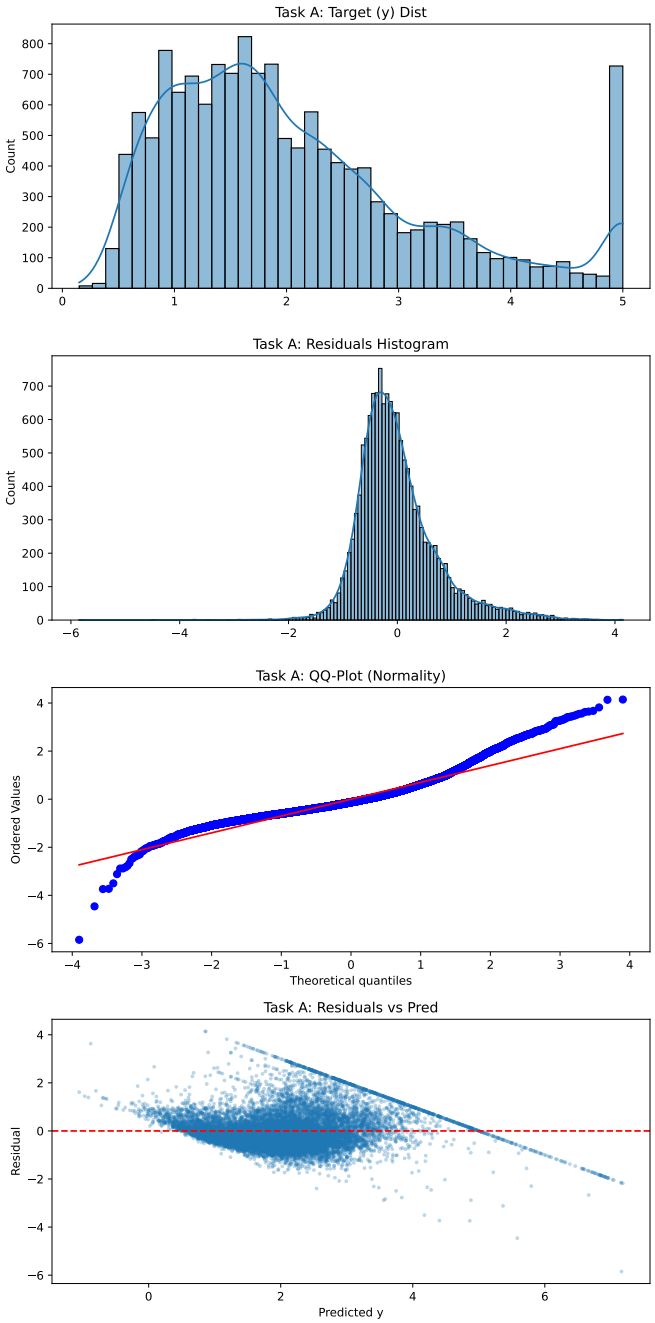


Figure 1: Diagnostics of task A.

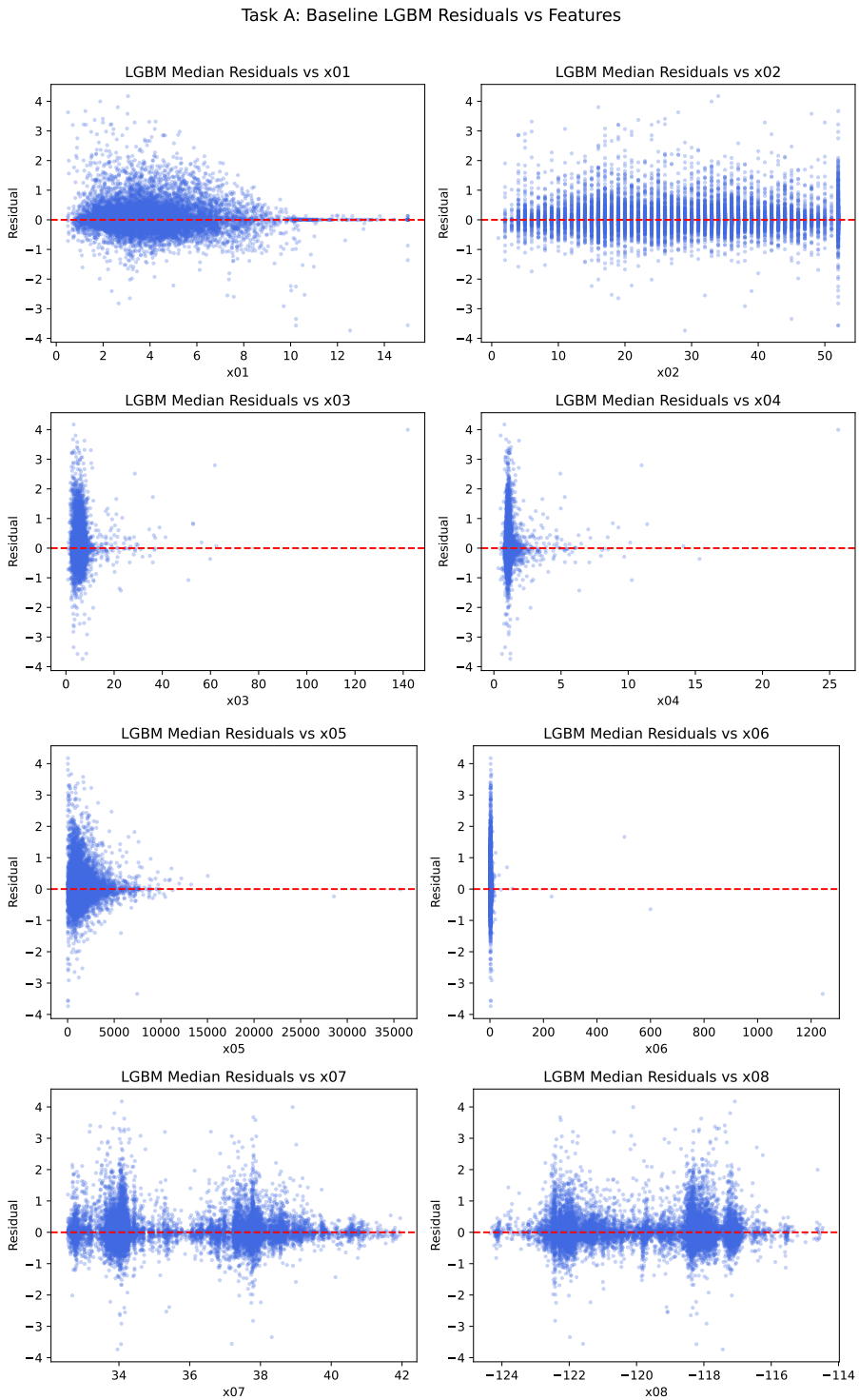


Figure 2: Residuals vs. features for task A.

The standout feature of task A is right-censoredness. This is evident from the clipped target at $y = 5$, visualized by the clear $5 - \hat{y}$ line in the residuals vs. prediction graph, which is also supported by the peak at $y = 5$ in the target distribution histogram. Initially, a `LGBM` model equipped with clipping final predictions at 5 was trained, reaching D^2-scores of 0.66. However, the decision was made to push the model a bit further by analyzing the residuals vs. features plots for the `LGBM` median. The plots of both $x07$ and $x08$ exhibited localized distinct clusters of high variance. The uniqueness of the pattern suggested that the features are related. As `LGBM` is inefficient at approximating diagonal or non-rectangular boundaries, the hypothesis was made that the features may represent some 2D space. This was addressed by using K-Means clustering with $k = 50$ to transform and replace the features with a 50-dimensional distance matrix to the cluster centers. All in all, this allowed for the `LGBM` to perform more efficiently, yielding a local validation D^2-score of 0.7. The finalized parameters of the model were optimized by Optuna to yield a local 0.7014 D^2-score, consistent with the Kaggle public score taking other tasks into account.

3 Task B

The defining characteristic of task B was multi-modality. Figure 3 makes this apparent, specifically the residuals histogram and the residuals vs. prediction scatter plot.

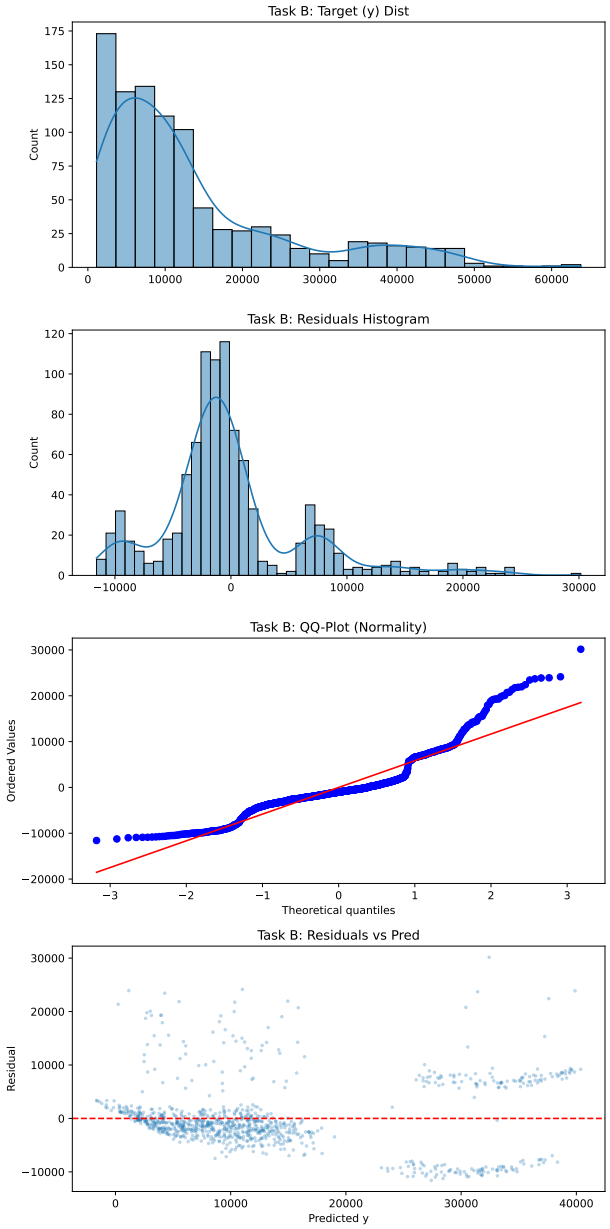


Figure 3: Diagnostics of task B.

The residuals vs. prediction plot exhibits distinct clusters of residual groups, strongly suggesting the presence of different regimes in the data. As the OLS baseline constantly predicted a middle-ground average, the resulting residuals for this multi-modal dataset appear in regimes. This observation is supported by the peaks in different groups of residuals, and by the S-shape in the QQ-plot.

However, the key limitation of task B was the limited size of the training data (only 937 training observations). Failing to account for this in training the ML model would result in severe overfitting. To assist the model in distinguishing between the modalities, the feature engineering had to take this into account. Instead of applying more aggressive techniques like K-Means clustering, a simpler Logistic Regression model was introduced to tackle this issue without adding unnecessary complexity. The role of the model was to predict the probability of an observation belonging to either half of the training data, based on the median of the target variable. Adding the probability as a meta-feature assisted the ML model in distinguishing between the modes.

The algorithm chosen for task B was `XGBoost`, after it showed better performance on the local split than `LGBM` and is less prone to overfitting on a small dataset. This is due to `LGBM`:s leaf-wise tree growth, while `XGBoost` takes a level-wise approach. `XGBoost` was equipped with robust regularization to prevent overfitting, and Optuna was used to optimize parameters (`max_depth=4, reg_lambda=0.5890`). The model successfully leveraged the meta-feature (better performance than without it) to yield a strong 0.7695 D^2-score on the local split, aligning closely with the Kaggle score.

4 Task C

In task C, the heteroscedastic nature of the data, displayed in Figure 4, was utilized to facilitate effective modeling.

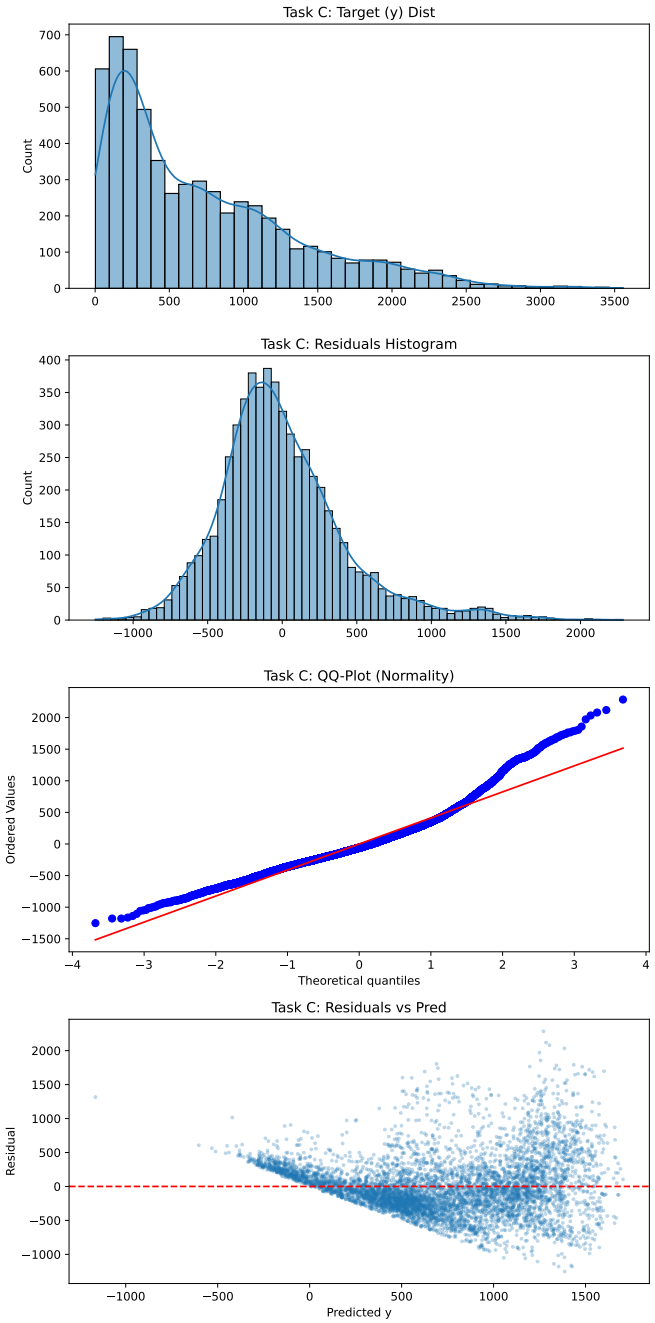


Figure 4: Diagnostics of task C

The core problem the OLS baseline fails to capture is heteroskedasticity, in the form of variance in the residuals increasing as the predicted $\hat{y}$ increases. The funnel-shape in the residuals vs. pred demonstrates this clearly.

The method chosen to solve this issue, after proving to perform the best, was an `LGBM` model equipped with the baseline OLS-model predictions as a meta-feature. This allowed the algorithm to dedicate all capacity to learning the heteroskedasticity of the data instead of wasting capacity on figuring out where the central line was. Interestingly, this approach was most efficient despite another similar `LGBM` engineered with a variance model meta feature was tested with local D^2 results around 0.76. The final version, which figured out the variance pattern on its own, reached a significantly better score of 0.8141.

5 Task D

The defining characteristics of task D were censoring combined with regime complexity. As the target distribution histogram in Figure 5 illustrates, there is a spike of values at $y = 2$, alongside a wide "hill" between $y = 10$ and $y = 20$. Furthermore, the lines $21 - \hat{y}$ and $-\hat{y}$ in the residuals vs. prediction plot indicate boundedness between targets 0 and 21.

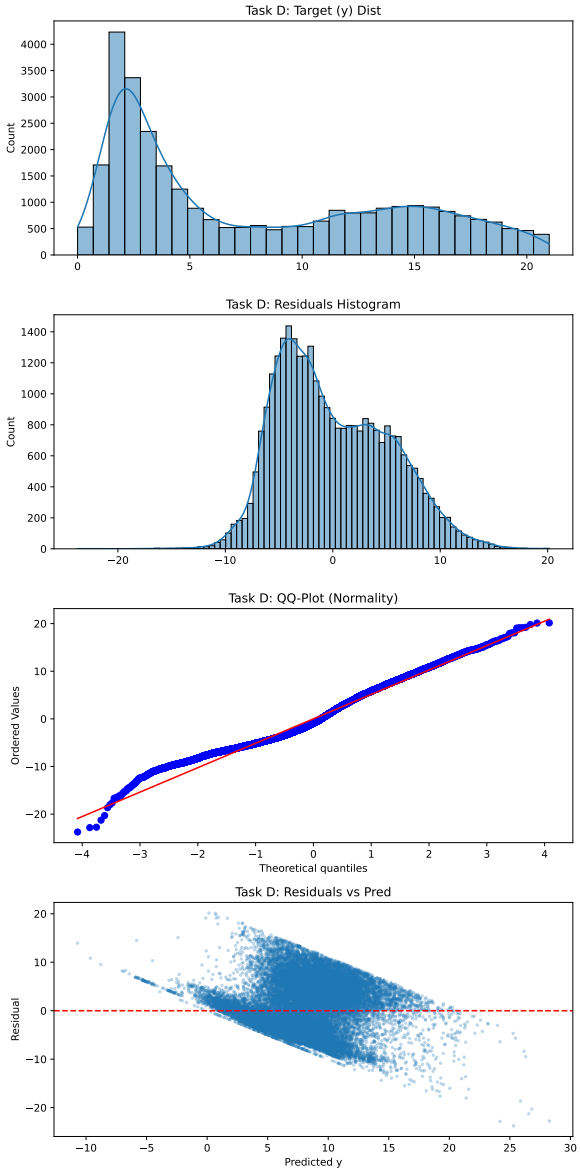


Figure 5: Diagnostics of task D.

When a single, global model (such as a standard `LGBM` regressor) attempts to minimize the Pinball loss across this dataset, it is forced to compromise between these different regimes[1]. Consequently, baseline models stagnated at a local D^2-score of approximately 0.46, struggling to simultaneously predict the exact left-censored boundary at zero and the high-variance spread of the upper hill.

To overcome this bottleneck, a "Mixture of Experts" architecture was designed. First, a threshold at $y = 6.9$ was established to separate the low and high regimes. A gating network, utilizing a calibrated `LGBMClassifier`, was trained to predict the probability that an unseen observation belongs to the high regime. The training data was then partitioned, and two specialized `LGBM` regressors were trained: a "low expert" focused entirely on the left-censored lower boundary and a "high expert" dedicated to the upper variance.

Because the final prediction requires a precise evaluation of the target's distribution, both experts were configured to predict 13 dense quantiles (from 1% to 99%) to construct a full CDF for every observation. The gating network's predicted probability was then used to weight and merge the two experts' CDFs into a single combined distribution. Finally, this aggregated CDF was inverted via interpolation to extract the exact nine quantiles for the Kaggle submission, with the final predictions clipped to respect the $[0, 21]$ boundary.

Since the two experts handled different types of data complexity, separate Optuna optimizations were run to find own hyperparameters for each expert. This tailored architecture improved upon the performance plateau, elevating the local validation D^2-score to 0.4711.

[1]AI Usage Declaration: The text in this section was drafted and refined with the assistance of Google Gemini, while the analytical observations and model architecture were independently directed and verified by the author.

6 Conclusion

In summary, this assignment demonstrated that direct quantile optimization using gradient boosting frameworks successfully captures complex, nonlinear relationships and severe departures from the Gaussian assumption. By tailoring the architecture and feature engineering to the specific characteristics of each dataset—the limitations of traditional parametric distributions were overcome.

For the final submission, the individual models for all four tasks were trained on the entirety of their respective training datasets to maximize data utilization. The combined final Kaggle submission yielded an overall public leaderboard score of 0.711. This result strongly aligns with the internal validation metrics observed during development, indicating that the implemented strategies, coupled with strict Optuna regularization, successfully generalized to the unseen test data without overfitting.